

Giraud Mirco

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Software Engineer | C++ · C#/.NET · Quantitative & Trading Systems

Software engineer with 4+ years building pricing, risk, and trading systems in C#/.NET within a €180B banking group; interest-rate risk, initial margin, and front-office tooling used daily by traders and risk teams. Now focused on low-latency C++: building itch-engine, a NASDAQ TotalView-ITCH 5.0 feed handler in C++23 with full L3 order-book reconstruction. Strong grounding in modern C++, concurrency, and performance optimization. Seeking a low-latency/systems C++ role; open to immediate relocation.

EMPLOYMENT EXPERIENCE

Iccrea Banking Group - Quantitative Developer

April 2022 – Today

Rome, Italy

- Designed and developed PMRM, the bank's official interest rate risk monitoring suite, used daily by front office, treasury, and risk teams to oversee exposures across multiple portfolios and derivative products while streamlining risk control processes.
- Developed C#/.NET and Python pricing and data-processing modules for Murex and FairMat, including ETL pipelines and certificate pricing implementing Jarrow-Yildirim and Heston models with a custom uplift module.
- Collaborated with front office traders to refine pricing logic and ensure accurate P&L attribution and sensitivities, porting legacy Excel/VBA add-ins into custom C# front-office tools for fast scenario testing.
- Built and optimized .NET engines for fixed income valuation, yield curves, and initial margin calculations, achieving up to 85% faster processing.

Selected Projects

- Portfolio Market Risk Monitor* (Risk Sensitivities and Derivatives): Developed a suite enabling granular analysis by portfolio, asset type, curve, and tenor; supports multiple currencies and ISIN-level drilldowns with a robust interactive visualization engine.
- Credit Risk Engine*: Built a C#/.NET module to process 119k+ credit risk records in 13 seconds, enabling real-time portfolio monitoring and stress testing.
- Derivatives Pricing Tool*: Multi-threaded valuation engine for swaps and bonds in Python + C#, cutting execution time by 85% and improving daily PnL reporting accuracy.

HRC srl. - Junior Developer

May 2021 – March 2022

Turin, Italy

- Improved prototype of GSS (Safety and Health Manager) web application for large industries (Fincantieri, Henkel, Epta), enhancing HSE data collection and reporting.
- Optimized ASP.NET MVC backend with Entity Framework and SQL Server integration to support high-volume HSE data capture and analysis.

OPEN-SOURCE PROJECTS

itch-engine - NASDAQ TotalView-ITCH 5.0 Feed Handler (C++23) · github.com/mircofx/itch-engine

- Low-latency feed handler decoding the full NASDAQ TotalView-ITCH 5.0 protocol in modern C++23; zero-copy ingestion via mmap, packed-struct overlays, and lazy std::byteswap.
- Full L3 limit-order-book reconstruction (add/cancel/execute/replace/delete) with per-symbol price-level and order-ID indexing, demultiplexing one interleaved stream into thousands of live per-instrument books.
- Performance-engineered book path: cache-conscious layout, fixed-point prices, flat price-ladder levels, and a flat open-addressed order table (−29% instructions, −40% cache misses vs std::unordered_map, ranked on hardware perf counters); lock-free SPSC queue splitting parse from book across two threads.
- Sustains 3.8 M msg/s single-threaded parse + L3 build (258 M msg/s framing ceiling) over a 64.9 M-message session, validated against an independent oracle book achieving 0 mismatches, no crossed books.

ADDITIONAL INFORMATION

Technical Skills

- C++: Modern C++ (C++11/23), low-latency & lock-free programming, cache optimization, zero-copy/mmap I/O, SPSC queues, micro-benchmarking
- C#/.NET: C# (.NET, .NET Core, MVC), multithreading, performance optimization, gRPC
- Other: Python, SQL, Git, Bloomberg API
- Financial Platforms & APIs: Bloomberg API, Finastra Summit, market data feeds, Excel/VBA
- Databases & Data Processing: SQL Server, relational databases, data integration & ETL pipelines
- Software Engineering: Object-oriented design, multi-threading, performance optimization, automated testing, version control (Git)
- Quantitative Finance: Derivatives pricing, credit risk models, yield curve construction, portfolio risk monitoring, initial margin & regulatory calculations

EDUCATION

Naples Fintech Laboratory (NFL) - Financial Innovation Lab

Naples, Italy | 2024

- Developed AI solutions for banking sector, including fraud detection and tailored communication tools based on GenAI

University of Turin - Economics & Statistics for Enterprises

Turin, Italy | 2019–2023

- Relevant modules: Financial Markets, Technical Analysis, Microeconomics

Polytechnic of Turin - Mechanical Engineering

Turin, Italy | 2017–2019

- English-taught course; exams in Mathematical Analysis I, Computer Science, English

Liceo Scientifico Carlo Cattaneo – Diploma in Applied Sciences

Turin, Italy | 2012–2017

- Ranked 1st in GioiaMathesis National Logical-Mathematical Olympiad